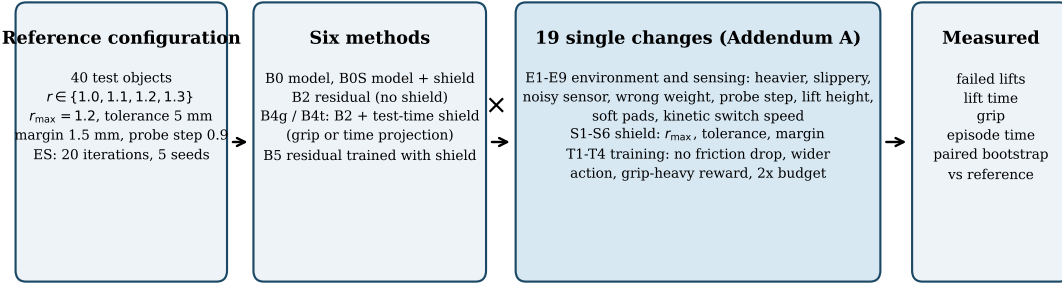
Reference configuration
40 test objects
$r \in \{1.0, 1.1, 1.2, 1.3\}$
$r_{max} = 1.2$, tolerance 5 mm
margin 1.5 mm, probe step 0.9
ES: 20 iterations, 5 seeds
Six methods
B0 model, B0S model + shield
B2 residual (no shield)
B4g / B4t: B2 + test-time shield
(grip or time projection)
B5 residual trained with shield
×
19 single changes (Addendum A)
E1-E9 environment and sensing: heavier, slippery,
noisy sensor, wrong weight, probe step, lift height,
soft pads, kinetic switch speed
S1-S6 shield: r_{max}, tolerance, margin
T1-T4 training: no friction drop, wider
action, grip-heavy reward, 2x budget
Measured
failed lifts
lift time
grip
episode time
paired bootstrap
vs reference